# Computation for φ³ Theory

*Source: Follows Srednicki and Peskin & Schroeder — counter-terms, partition functional, and one-loop diagrams in φ³.*

## 1 Computation for $\phi^3$ Theory

All of the calculation below is based on Srednicki, and Peskin and Schroeder, so nothing here is original at all.

Here, we will use the metric signature $(-,+,+,+)$. The Lagrangian of $\phi^3$ is given by

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 + \frac{1}{3!}g\phi^3. \tag{1.1}$$

To make the theory satisfying

$$\langle k|\phi(x)|\Omega\rangle = e^{-ik\cdot x}, \quad \langle\Omega|\phi(x)|\Omega\rangle = 0, \tag{1.2}$$

we make field redefinitions so that the Lagrangian becomes

$$\mathcal{L} = -\frac{1}{2}Z_\phi\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}Z_m m^2\phi^2 + \frac{1}{3!}Z_g g\phi^3 + Y\phi = \mathcal{L}_0 + \mathcal{L}_1, \tag{1.3}$$

where

$$\mathcal{L}_0 = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2, \tag{1.4}$$

and

$$\mathcal{L}_1 = -\frac{1}{2}(Z_\phi - 1)\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}(Z_m - 1)m^2\phi^2 + \frac{1}{3!}Z_g g\phi^3 + Y\phi. \tag{1.5}$$

The partition functional can be written as

$$Z = \int \mathcal{D}\phi(x)e^{i\int d^dx(\mathcal{L}_0+J\phi+\mathcal{L}_1)} = e^{i\int d^dx\mathcal{L}_1(-i\delta/\delta J(x))}\int \mathcal{D}\phi(x)e^{i\int d^dx(\mathcal{L}_0+J\phi)} = e^{i\int d^dx\mathcal{L}_1(-i\delta/\delta J(x))}Z_0, \tag{1.6}$$

where

$$Z_0 = \int \mathcal{D}\phi(x)e^{i\int d^dx(\mathcal{L}_0+J\phi)}. \tag{1.7}$$

Making the field redefinition

$$\chi(x) = \phi(x) - iD_F(x-y)J(y), \tag{1.8}$$

where

$$(\partial^2 - m^2)D_F(x-y) = i\delta^d(x-y). \tag{1.9}$$

Note that since the above is just a shift of field, the measure is unchanged $\mathcal{D}\phi = \mathcal{D}\chi$. Then, the integrand of $Z_0$ becomes

$$\exp\left[i\int d^dx\left(\frac{1}{2}\chi(x)(\partial^2-m^2)\chi(x)+\frac{i}{2}\int d^dyJ(x)D_F(x-y)J(y)\right)\right], \tag{1.10}$$

and hence

$$Z_0=\mathcal{N}\exp\left[-\frac{1}{2}\int d^dx\int d^dyJ(x)D_F(x-y)J(y)\right], \tag{1.11}$$

where $\mathcal{N}$ is formally infinite, but since it is not a measurable quantity, it does not boarder us. Now the full partition functional becomes

$$Z=e^{i\int d^dx\mathcal{L}_1(-i\delta/\delta J(x))}\exp\left[-\frac{1}{2}\int d^dx\int d^dyJ(x)D_F(x-y)J(y)\right] \tag{1.12}$$

up to some irrelevant constant.

Now, we are interested in finding the exact two-point correlation function to one-loop order. That is

$$\langle\Omega|T\phi(x_1)\phi(x_2)|\Omega\rangle=(-i)^2\frac{\delta^2}{\delta J(x_1)\delta J(x_2)}\ln Z[J]\bigg|_{J=0}=i(-i)^2\frac{\delta^2}{\delta J(x_1)\delta J(x_2)}W[J]\bigg|_{J=0}, \tag{1.13}$$

where $W[J]$ is the connected diagram generating functional

$$Z[J]\equiv e^{iW[J]}. \tag{1.14}$$

In practice, one can simply write down the Feynman rules and compute the correlation functions. However, for our purpose, we shall compute it directly from functional derivatives. We shall first compute $Z[J]$:

$$\begin{aligned}Z[J]&=\left[1+i\int d^dx\frac{1}{2}\left(-i\frac{\delta}{\delta J(x)}\right)(A\partial_x^2-Bm^2)\left(-i\frac{\delta}{\delta J(x)}\right)+\cdots\right]Z_0\\&=Z_0-\frac{i}{2}\int d^dx\left(-i\frac{\delta}{\delta J(x)}\right)(A\partial_x^2-Bm^2)\left(-\int d^dyJ(y)D_F(x-y)\right)Z_0+\cdots\\&=Z_0-\frac{i}{2}\int d^dx\left(\int d^dzJ(z)D_F(x-z)\right)(A\partial_x^2-Bm^2)\left(\int d^dyJ(y)D_F(x-y)\right)Z_0+\cdots,\end{aligned} \tag{1.15}$$

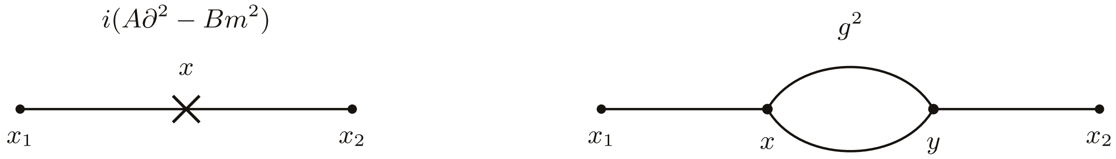
where we have focused on the connected two-point term since we will eventually take $J=0$ after the functional derivative. However, one should not forget the other term

$$\begin{aligned}&\frac{i^2}{(3!)^2}g^2\int d^dx\int d^dy\left(-i\frac{\delta}{\delta J(x)}\right)^3\left(-i\frac{\delta}{\delta J(y)}\right)^3Z_0\\&=-\frac{g^2}{2}\int d^dx\int d^dyJ(x)D_F(x-y)^2J(y)Z_0+\cdots\end{aligned} \tag{1.16}$$

Now, we can compute the two-point correlation function; that is

$$\begin{aligned}D(x_1-x_2)=\langle\Omega|T\phi(x_1)\phi(x_2)|\Omega\rangle&=-\frac{\delta^2}{\delta J(x_1)\delta J(x_2)}Z[J]\bigg|_{J=0}\\&=D_F(x_1-x_2)+i\int d^dxD_F(x-x_1)(A\partial_x^2-Bm^2)D_F(x-x_2)\\&\quad-\frac{g^2}{2}\int d^dx\int d^dyD_F(x_1-x)D_F(x-y)^2D_F(y-x_2)+\mathcal{O}(g^4),\end{aligned} \tag{1.17}$$

where we have been quite sloppy with the overall constant throughout. These are exactly the diagrams one would expect from Feynman rules




(a) (b)

In momentum space, we have

$$\tilde{D}(k^2) = \tilde{D}(k^2) + \tilde{D}_F(k^2) i\Pi(k^2)\tilde{D}_F(k^2) + \mathcal{O}(g^4), \tag{1.18}$$

where

$$i\Pi(k^2) = -i(Ak^2 + Bm^2) - \frac{g^2}{2}\int \frac{d^d l}{(2\pi)^d}\frac{i}{l^2+m^2-i\epsilon}\frac{i}{(k+l)^2+m^2-i\epsilon} + \mathcal{O}(g^4). \tag{1.19}$$

Consider the con exact propagator, where $\Pi(k^2)$ is now to all order of $g$,

$$\tilde{D}(k^2) = \tilde{D}(k^2)\sum_{n=0}^{\infty}[i\Pi(k^2)\tilde{D}_F(k^2)]^2 = \frac{i}{k^2+m^2-i\epsilon-\Pi(k^2)}. \tag{1.20}$$

For the particle to have mass $m$, we need the exact propagator with residue one at $k^2 = -m^2$. Therefore, we require

$$\Pi(-m^2) = \Pi'(-m^2) = 0, \tag{1.21}$$

which helps us fix $A$ and $B$. Now, we need to compute the integral above.

Now, we shall perform the integral in $d$ dimensions. To do that, we use the Feynman's formula.

**Claim:**

$$\prod_{j=1}^{n}\frac{1}{A_j^{\alpha_j}} = \frac{\Gamma(\sum_i \alpha_i)}{\prod_i \Gamma(\alpha_i)}\frac{1}{(n-1)!}\int dF_n \frac{\prod_i x_i^{\alpha_i - 1}}{(\sum_i x_i A_i)^{\sum_i \alpha_i}}, \tag{1.22}$$

where

$$dF_n = (n-1)!\int_0^1 dx_1 \cdots \int_0^1 dx_n \delta(1 - \sum_i x_i). \tag{1.23}$$

**Proof:** Consider

$$\frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} = \int_0^{\infty} dt_i\, t_i^{\alpha_i - 1} e^{-A_i t_i}. \tag{1.24}$$

Then, we take

$$\prod_{i=1}^{n}\frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} = \prod_{i=1}^{n}\int_0^{\infty} dt_i\, t_i^{\alpha_i - 1} e^{-A_i t_i}. \tag{1.25}$$

We can multiply the right-hand side by

$$1 = \int_0^{\infty} ds\, \delta(s - \sum_i t_i), \tag{1.26}$$

which gives

$$\prod_{i=1}^{n} \frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} = \int_0^\infty ds\, \delta(s - \sum_i t_i) \prod_{i=1}^{n} \int_0^\infty dt_i\, t_i^{\alpha_i - 1} e^{-A_i t_i}. \tag{1.27}$$

Now, making the substitution $t_i = s x_i$, we find

$$\begin{aligned} \prod_{i=1}^{n} \frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} &= \int_0^\infty ds\, \delta(s - s\sum_i x_i) \prod_{i=1}^{n} \int_0^\infty dx_i\, s^{\alpha_i} x_i^{\alpha_i - 1} e^{-sA_i x_i} \\ &= \frac{1}{(n-1)!} \int dF_n \left( \prod_{i=1}^{n} x_i^{\alpha_i - 1} \right) \int_0^\infty ds\, s^{\sum_i \alpha_i - 1} e^{-\sum_i s A_i x_i} \\ &= \frac{1}{(n-1)!} \int dF_n \left( \prod_{i=1}^{n} x_i^{\alpha_i - 1} \right) \frac{\Gamma(\sum_i \alpha_i)}{(\sum_i A_i x_i)^{\sum_i \alpha_i}}. \end{aligned} \tag{1.28}$$

We use ∎

$$\frac{1}{A_1 A_2} = \int_0^1 dx_1 \int_0^1 dx_2\, \delta(x_1 + x_2 - 1) \frac{1}{(x_1 A_1 + x_2 A_2)^2} = \int_0^1 dx \frac{1}{[xA_1 + (1-x)A_2]^2}. \tag{1.29}$$

Therefore, we have

$$\begin{aligned} \frac{1}{l^2 + m^2} \frac{1}{(k+l)^2 + m^2} &= \int_0^1 dx [(1-x)(l^2 + m^2) + x((k+l)^2 + m^2)]^{-2} \\ &= \int_0^1 dx [l^2 + 2xkl + m^2 + xk^2]^{-2} \\ &= \int_0^1 dx [(l + xk)^2 + m^2 + x(1-x)k^2]^{-2}, \end{aligned} \tag{1.30}$$

where we have abbreviated $m^2 - i\epsilon$ as $m^2$. Now, we write $q \equiv l + xk$ and $D \equiv m^2 + x(1-x)k^2$. So that the integral is now

$$-\frac{g^2}{2} \int \frac{d^d l}{(2\pi)^d} \frac{i}{l^2 + m^2 - i\epsilon} \frac{i}{(k+l)^2 + m^2 - i\epsilon} = \frac{g^2}{2} \int_0^1 dx \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + D)^2}. \tag{1.31}$$

Then, we perform the Wick rotation so that $q^0 = i q_E^0$, $\mathbf{q} = \mathbf{q}_E$, $q^2 = q_E^2$, and $d^d q = i d^d q_E$. Therefore, the above integral becomes

$$I(k^2) = i \int_0^1 dx \int \frac{d^d q_E}{(2\pi)^d} \frac{1}{(q_E^2 + D)^2}, \tag{1.32}$$

where

$$i\Pi(k^2) = \frac{g^2}{2} I(k^2) - i(Ak^2 + Bm^2) + \mathcal{O}(g^4). \tag{1.33}$$

With a slight abbreviation of notation, we now write the integral measure as $d^d q_E = q_E^{d-1} dq_E d\Omega_{d-1}$, where $q_E$ is now the length of the vector $q_E$.

**Claim:**

$$\Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}. \tag{1.34}$$

**Proof:** Consider the integral

$$\int_{\mathbb{R}^d} d^d x\, e^{-x^2} = \int d\Omega_{d-1} \int_0^\infty dr\, r^{d-1} e^{-r^2}. \tag{1.35}$$

The left-hand side gives

$$\left( \int_{\mathbb{R}} dx\, e^{-x^2} \right)^d = \pi^{d/2}, \tag{1.36}$$

and the right-hand side gives

$$\Omega_{d-1} \frac{1}{2} \int_0^\infty dr\, r^{d/2-1} e^{-r} = \Omega_{d-1} \frac{\Gamma(d/2)}{2}. \tag{1.37}$$

We shall also prove the general identity, that would turn out to be useful in many cases when dealing with loop corrections. ∎

**Claim:**

$$\int \frac{d^d q_E}{(2\pi)^d} \frac{(q_E^2)^n}{(q_E^2 + D)^m} = \frac{\Gamma(m-n-d/2)\Gamma(n+d/2)}{(4\pi)^{d/2}\Gamma(m)\Gamma(d/2)} D^{-(m-n-d/2)}. \tag{1.38}$$

**Proof:**

$$\int \frac{d^d q_E}{(2\pi)^d} \frac{(q_E^2)^n}{(q_E^2 + D)^m} = \frac{2\pi^{d/2}}{\Gamma(d/2)} \frac{1}{(2\pi)^d} \int_0^\infty dq_E \frac{q_E^{2n+d-1}}{(q_E^2 + D)^m}. \tag{1.39}$$

Making the substitution $s^2 = q^2/D + 1$, we find

$$\int \frac{d^d q_E}{(2\pi)^d} \frac{(q_E^2)^n}{(q_E^2 + D)^m} = \frac{2D^{n+d-m-1}}{(4\pi)^{d/2}\Gamma(d/2)} \int_1^\infty ds \frac{(s^2-1)^{n+d/2-1}}{s^{2m-1}}. \tag{1.40}$$

Then, we substitute $t = 1/s^2$, we find

$$\begin{aligned}
\int \frac{d^d q_E}{(2\pi)^d} \frac{(q_E^2)^n}{(q_E^2 + D)^m} &= \frac{D^{n+d-m}}{(4\pi)^{d/2}\Gamma(d/2)} \int_0^1 dt\, t^{m-n-d/2-1} (t-1)^{n+d/2-1} \\
&= \frac{D^{n+d-m}}{(4\pi)^{d/2}\Gamma(d/2)} \beta(m-n-d/2, n+d/2-1) \\
&= \frac{D^{n+d-m}}{(4\pi)^{d/2}\Gamma(d/2)} \frac{\Gamma(m-n-d/2)\Gamma(n+d/2)}{\Gamma(m)}.
\end{aligned} \tag{1.41}$$

For our case, we have $n = 0$ and $m = 2$, and hence ∎

$$I(k^2) = i \int_0^1 dx \frac{\Gamma(2-d/2)}{(4\pi)^{d/2}} D^{-2+d/2}. \tag{1.42}$$

Now let us focus on the $\phi^3$ theory in $d = 6$, where $[g] = 0$ and the theory is renormalizable. For general dimensions, $[g] = 3 - d/2 \equiv \varepsilon/2$. We shall redefine the coupling constant to make it dimension less, that is

$$g \to g\tilde{\mu}^{\varepsilon/2}. \tag{1.43}$$

Now, we find

$$g^2\tilde{\mu}I(k^2) = ig^2\tilde{\mu}\int_0^1 dx \frac{\Gamma(\varepsilon/2-1)}{(4\pi)^{3-\varepsilon/2}} D^{-\varepsilon/2+1} = ig^2 \frac{\Gamma(-1+\varepsilon/2)}{(4\pi)^3}\int_0^1 dx\, D\left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\varepsilon/2}. \tag{1.44}$$

Using dimensional regularization, we take $\varepsilon \to 0$ so that

$$\Gamma(-1+\varepsilon/2) \sim -\frac{2}{\varepsilon} + \gamma - 1 + \cdots, \tag{1.45}$$

and

$$\left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\varepsilon/2} \sim 1 + \frac{\varepsilon}{2}\ln\left(\frac{4\pi\tilde{\mu}^2}{D}\right) + \cdots. \tag{1.46}$$

Then,

$$\Gamma(-1+\varepsilon/2)\left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\varepsilon/2} \sim -\frac{2}{\varepsilon} + \gamma - 1 - \ln\left(\frac{4\pi\tilde{\mu}^2}{D}\right) + \mathcal{O}(\varepsilon). \tag{1.47}$$

Defining

$$\mu = \sqrt{4\pi}e^{-\gamma/2}\tilde{\mu}, \quad \alpha = \frac{g^2}{(4\pi)^3} \tag{1.48}$$

we find

$$g^2\tilde{\mu}I(k^2) \sim i\alpha\int_0^1 dx\, D\left[-\frac{2}{\varepsilon} - 1 + \ln\left(\frac{m^2}{\mu^2}\right) - \ln\left(\frac{m^2}{D}\right) + \mathcal{O}(\varepsilon)\right]. \tag{1.49}$$

Now, we evaluate

$$\int_0^1 dx D = \int_0^1 dx(m^2 + x(1-x)k^2) = m^2 + k^2/6. \tag{1.50}$$

Therefore, we find

$$\begin{aligned}\Pi(k^2) &\sim \frac{\alpha}{2}\left[-(2/\varepsilon + 1 + 2\ln(\mu/m))(m^2+k^2/6) + \int_0^1 dx\, D\ln\big(D/m^2\big)\right] - (Ak^2 + Bm^2) + \mathcal{O}(\alpha^2) \\ &= \frac{\alpha}{2}\int_0^1 dx\, D\ln\big(D/m^2\big) - \left[\frac{1}{6}\alpha(\varepsilon + \ln(\mu/m) + 1/2) + A\right]k^2 \\ &\qquad\qquad - [\alpha(\varepsilon + \ln(\mu/m) + 1/2) + A]m^2 + \mathcal{O}(\alpha^2).\end{aligned} \tag{1.51}$$

To further simplify the expression, we shall let $A$ and $B$ to take the form

$$A = -\frac{1}{6}\alpha(\varepsilon + \ln(\mu/m) + 1/2) + \alpha\kappa_A - \mathcal{O}(\alpha^2), \tag{1.52}$$

$$B = -\alpha(\varepsilon + \ln(\mu/m) + 1/2) - \alpha\kappa_B + \mathcal{O}(\alpha^2). \tag{1.53}$$

Then, we find

$$\Pi(k^2) \sim \frac{\alpha}{2}\int_0^1 dx\, D\ln\big(D/m^2\big) - \alpha\left(\frac{1}{6}\kappa_A k^2 + \kappa_B m^2\right) + \mathcal{O}(\alpha^2). \tag{1.54}$$

Now, we impose the renormalization condition $\Pi(-m^2) = \Pi'(-m^2) = 0$ to fix $\kappa_A$ and $\kappa_B$. To do that, we note that $\Pi(-k^2)$ must take the form

$$\Pi(-k^2) \sim \frac{\alpha}{2}\int_0^1 dx\, D\ln(D/D_0) + c(k^2+m^2) + \mathcal{O}(\alpha^2), \tag{1.55}$$

where $c \in \mathbb{R}$ and

$$D_0 = D\big|_{k^2=-m^2} = [1-x(1-x)]m^2. \tag{1.56}$$

Then, we differentiate with respect to $k^2$ to find

$$\Pi'(-k^2) \sim \frac{\alpha}{2}\int_0^1 dx\, \left[x(1-x)\ln(D/D_0) + D_0 x(1-x)\right] + c + \mathcal{O}(\alpha^2). \tag{1.57}$$

Therefore,

$$\Pi'(-m^2) = 0 \implies c = -\frac{\alpha}{12}. \tag{1.58}$$

Finally we obtain the 1-loop correction for the two-point correlation function, which is

$$\Pi(-k^2) \sim \frac{\alpha}{2}\int_0^1 dx\, D\ln(D/D_0) - \frac{\alpha}{12}(k^2+m^2) + \mathcal{O}(\alpha^2). \tag{1.59}$$

We might as well right down the asymptotic value for $\kappa_A$ and $\kappa_B$. That is

$$\kappa_A = \frac{1}{4}, \quad \kappa_B = \frac{1}{24}. \tag{1.60}$$

Note that the integral in our final expression is real if $D > 0$. That is $x(1-x)k^2 > -m^2$, and since $x(1-x) \in [0, 1/4)$ we must have $k^2 > -4m^2$ for real $\Pi(k^2)$. Then, one might be tempting to ask: what happens if $k^2 < -4m^2$, and $\operatorname{Im}\Pi(k^2) \neq 0$? To understand what is happening, let us write the one-loop correction in Lehmann-Källén form,

$$\tilde{D}(k^2) = \frac{i}{k^2+m^2-i\epsilon} + \int_{4m^2}^{\infty} ds\, \rho(s)\frac{i}{k^2+s-i\epsilon} = \frac{i}{k^2+m^2-i\epsilon-\Pi(k^2)}. \tag{1.61}$$

Using the identity

$$\frac{1}{x-i\epsilon} = P\frac{1}{x} + i\pi\delta(x), \tag{1.62}$$

we find

$$\operatorname{Im}(-i\tilde{D}(k^2)) = \pi\delta(k^2+m^2) + \pi\rho(-k^2) = \pi\delta(k^2+m^2-\Pi(k^2)), \tag{1.63}$$

where $k^2 > -4m^2$. Therefore, for $k^2 > -4m^2$, $\rho(-k^2) = \rho(s) = 0$. However, for $k^2 < -4m^2$, $\operatorname{Im}\Pi(k^2) \neq 0$. Hence, one would find

$$\pi\rho(s) = \frac{\operatorname{Im}\Pi(-s)}{(-s+m^2+\operatorname{Re}\Pi(-s))^2 + (\operatorname{Im}\Pi(-s))^2}. \tag{1.64}$$

Now, let us compute the vertex correction for the $\phi^3$ theory. In the previous computation we took a slightly more complicated root by evaluating the path integral straight away. Now, let us go back and write down the momentum space vertex Feynman rules of the theory:

$$iZ_g g. \tag{1.65}$$

We can define an exact three-point vertex $iV_3(k_1, k_2, k_3)$, where $k_1 + k_2 + k_3 = 0$. To one-loop correction, the exact three-point is given by

$$iV_3(k_1, k_2, k_3)/g = iZ_g + i(ig)^2 \int \frac{d^d l}{(2\pi)^d} \tilde{D}_F(l^2)\tilde{D}_F((l-k_1)^2)\tilde{D}_F((l+k_2)^2) + \mathcal{O}(g^5), \tag{1.66}$$

where we have used the fact that $Z_g = 1 + \mathcal{O}(g^2)$. We shall impose the natural renormalization condition,

$$iV_3(0,0,0) = 0, \tag{1.67}$$

to fix the value of $Z_g$. Just like before, we use the Feynman integration formula to get

$$\begin{aligned}
&\tilde{D}_F(l^2)\tilde{D}_F((l-k_1)^2)\tilde{D}_F((l+k_2)^2) \\
=&i^3 \int dF_3 \frac{1}{[x_1(l^2+m^2) + x_2((l-k_1)^2+m^2) + x_3((l+k_2)^2+m^2)]^3} \\
=&-i \int dF_3 \frac{1}{[l^2 + m^2 - 2x_2 k_1 l + 2x_3 k_2 l + x_2 k_1^2 + x_3 k_2^2]^3} \\
=&-i \int dF_3 \frac{1}{[(l - x_2 k_1 + x_3 k_2)^2 + m^2 + x_2 k_1^2 + x_3 k_2^2]^3} \\
=&-i \int dF_3 \frac{1}{(q^2+D)^3},
\end{aligned} \tag{1.68}$$

where $q \equiv l - x_2 k_1 + x_3 k_2$ and $D \equiv m^2 + x_2 k_1^2 + x_3 k_2^2$. Then, we have

$$\int \frac{d^d l}{(2\pi)^d} \tilde{D}_F(l^2)\tilde{D}_F((l-k_1)^2)\tilde{D}_F((l+k_2)^2) = -i \int dF_3 \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2+D)^3}. \tag{1.69}$$

Perform a Wick rotation, we find

$$\begin{aligned}
-i \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2+D)^3} &= \int \frac{d^d q_E}{(2\pi)^d} \frac{1}{(q_E^2+D)^3} \\
&= \frac{\Gamma(3-d/2)}{2(4\pi)^{d/2}} D^{-3+d/2}.
\end{aligned} \tag{1.70}$$

Now, substitute $\varepsilon = 6 - d$ and redefine the coupling constant just as before. We look for the asymptotic series of $\varepsilon \to 0$. That is

$$\begin{aligned}
-i(i^3 g^2)\tilde{\mu}^\varepsilon \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2+D)^3} &= \frac{ig^2}{2(4\pi)^3} \Gamma(\varepsilon/2) \left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\varepsilon/2} \\
&\sim \frac{ig^2}{2(4\pi)^3} (2/\varepsilon - \gamma + \cdots) \left(1 + \frac{\varepsilon}{2} \ln\left(\frac{4\pi\tilde{\mu}^2}{D}\right) + \cdots\right) \\
&= \frac{ig^2}{2(4\pi)^3} \left(\frac{2}{\varepsilon} - \gamma + \ln\left(\frac{4\pi\tilde{\mu}^2}{D}\right) + \cdots\right) \\
&= \frac{i\alpha}{2} \left(\frac{2}{\varepsilon} + \ln\left(\frac{\mu^2}{D}\right) + \cdots\right),
\end{aligned} \tag{1.71}$$

where we have used $\mu^2 = 4\pi e^{-\gamma}\tilde{\mu}^2$ and $\alpha \equiv g^2/(4\pi)^3$. Finally, we find the three-point vertex to be

$$iV_3(k_1, k_2, k_3)/g \sim iZ_g + i\frac{\alpha}{2}\left(\frac{2}{\varepsilon} + \int dF_3 \ln(\mu^2/D) + \cdots\right) + \mathcal{O}(g^5). \tag{1.72}$$

By setting $Z_g = 1 + C$, we find

$$V_3(k_1,k_2,k_3)/g \sim 1+\alpha\left[\frac{1}{\varepsilon}+\ln(\mu/m)+\frac{1}{2}\int dF_3\ln\bigl(m^2/D\bigr)+\ldots\right]+C+\mathcal{O}(\alpha^2). \tag{1.73}$$

Again, we let $C$ to take the form

$$C=-\alpha\left[\frac{1}{\varepsilon}+\ln(\mu/m)+\kappa_C\right]+\mathcal{O}(\alpha^2), \tag{1.74}$$

which gives

$$V_3(k_1,k_2,k_3)/g \sim 1-\frac{\alpha}{2}\int dF_3\ln\bigl(D/m^2\bigr)-\alpha\kappa_C+\mathcal{O}(\alpha^2). \tag{1.75}$$

Now, we impose the renormalization condition, $V_3(0,0,0)=0$, which gives $D=m^2$, and hence

$$\kappa_C=0. \tag{1.76}$$

Therefore, our final expression for the three-point vertex to first order in $\alpha$ is given by

$$V_3(k_1,k_2,k_3)/g \sim 1-\frac{\alpha}{2}\int dF_3\ln\bigl(D/m^2\bigr)C+\mathcal{O}(\alpha^2). \tag{1.77}$$

Now, we have determined all three of the renormalization factor to one-loop order.